

Week 01 Task 3

Lab Setup VirtualBox + Kali/Ubuntu + Windows VM

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Submitted To: TechBiz Security

1 Executive Summary

In the field of cyber security particularly in ethical hacking. It is very Important to work in safe and isolated network environment before initiating a security assessment against any target. This is important to avoid unintended attacks on live production system other than that this is also important when dealing with malware analysis this isolation will help you to prevent your network and data from malicious actors this report present how to setup a professional and secure lab environment for ethical hacking.

2 Scope of Task

The Scope of this task is to build a safe virtual environment for penetration testing. Install and Configure Oracle virtual box. Import and run the kali linux virtual machine after then verify that the system and network is correctly working

3 Tools Used

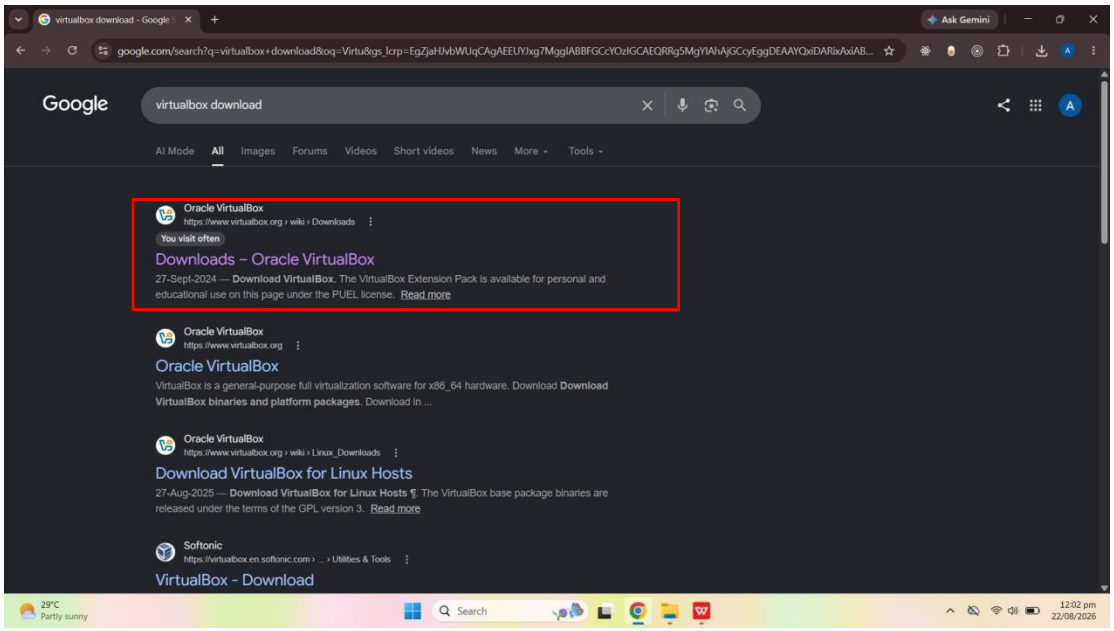
1. Virtual box
2. windows 11
3. Kali Linux (Attacker Machine)

4 Downloading and Installing Virtual box

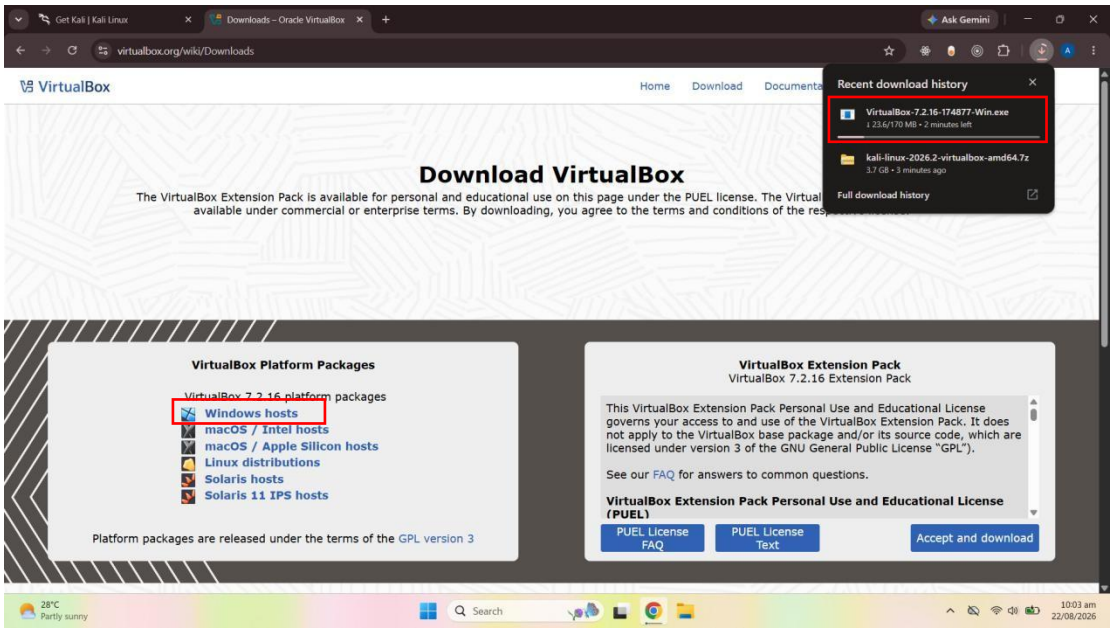
4.1 Downloading Virtual box

To Download the virtual box follow the steps given below

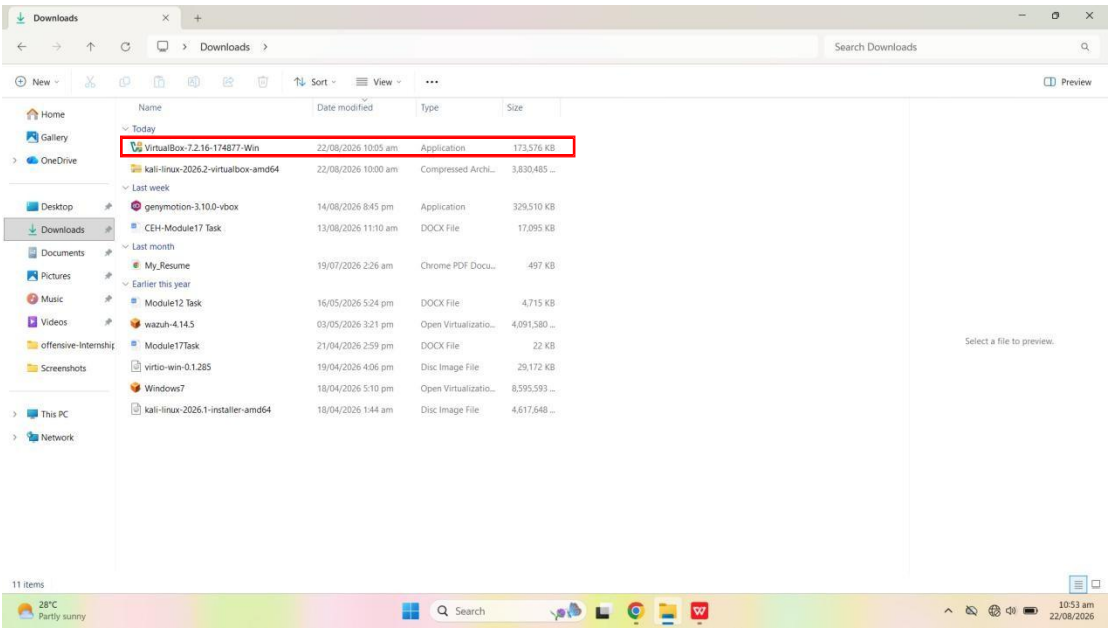
1) Go to Google.com and Search Virtual box Download select the official website Link: https://www.virtualbox.org



2) Open a website Select the virtual box download according to your host operating system as my Host operating system is windows so I selected windows.



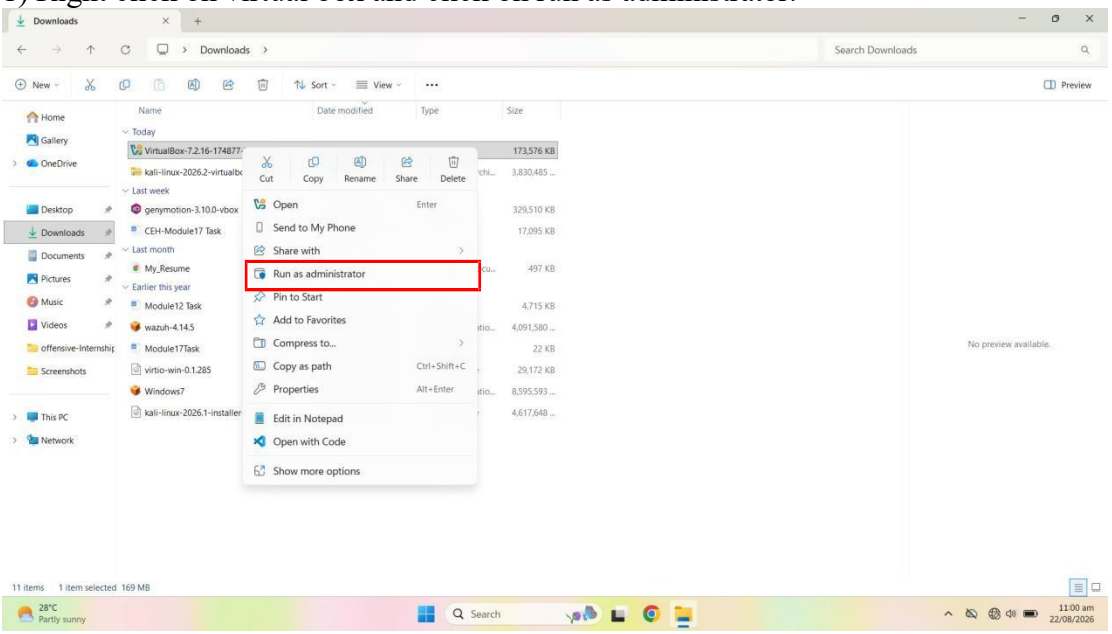
3) As Virtual box is successfully Downloaded can be seen in Downloads folder.



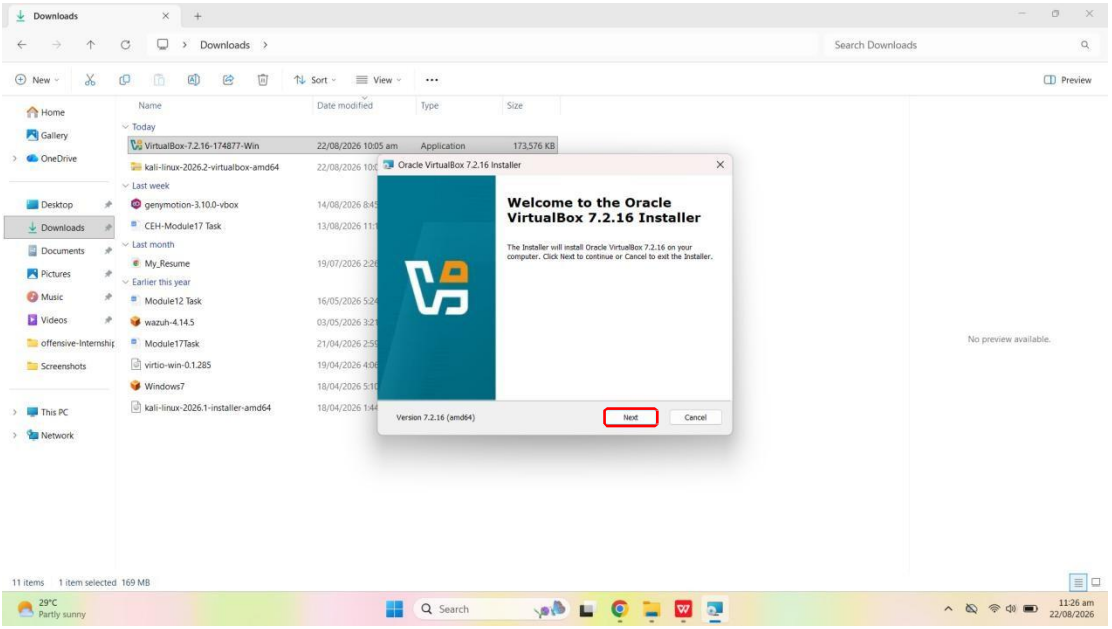
4.2 Installing Virtual box

After downloading follow the steps given below to install virtual box in a System.

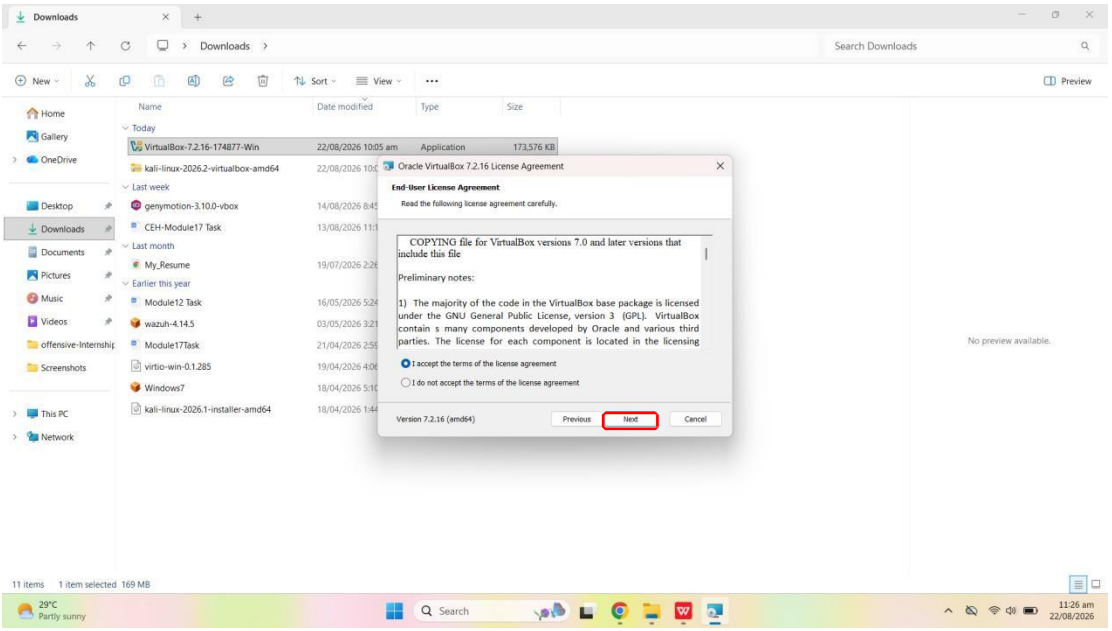
1) Right click on virtual box and click on run as administrator.



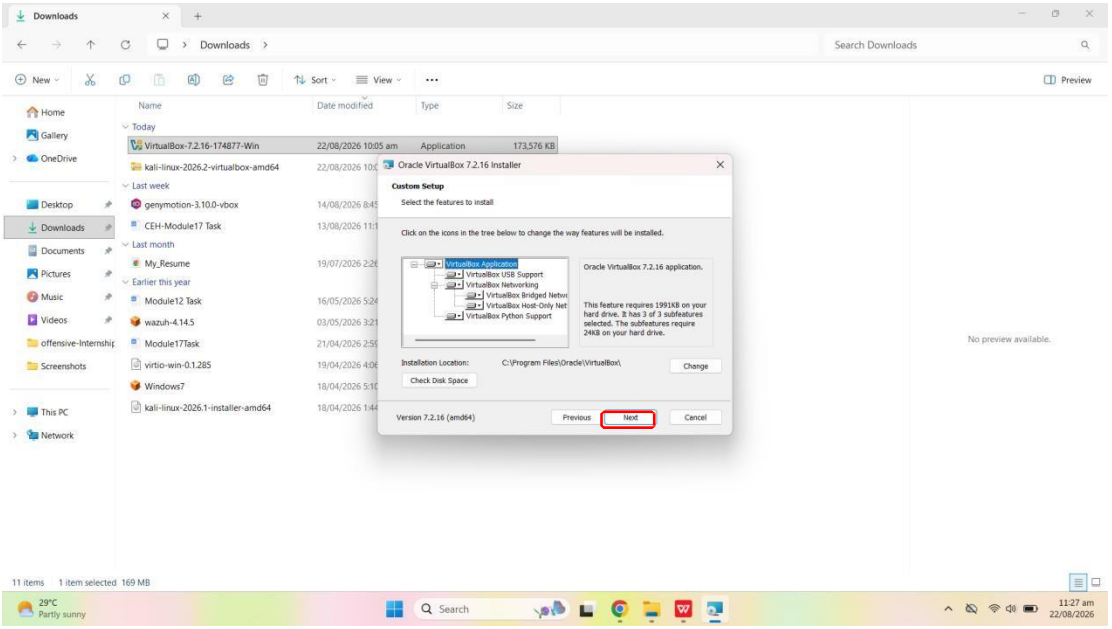
2) Click on Next



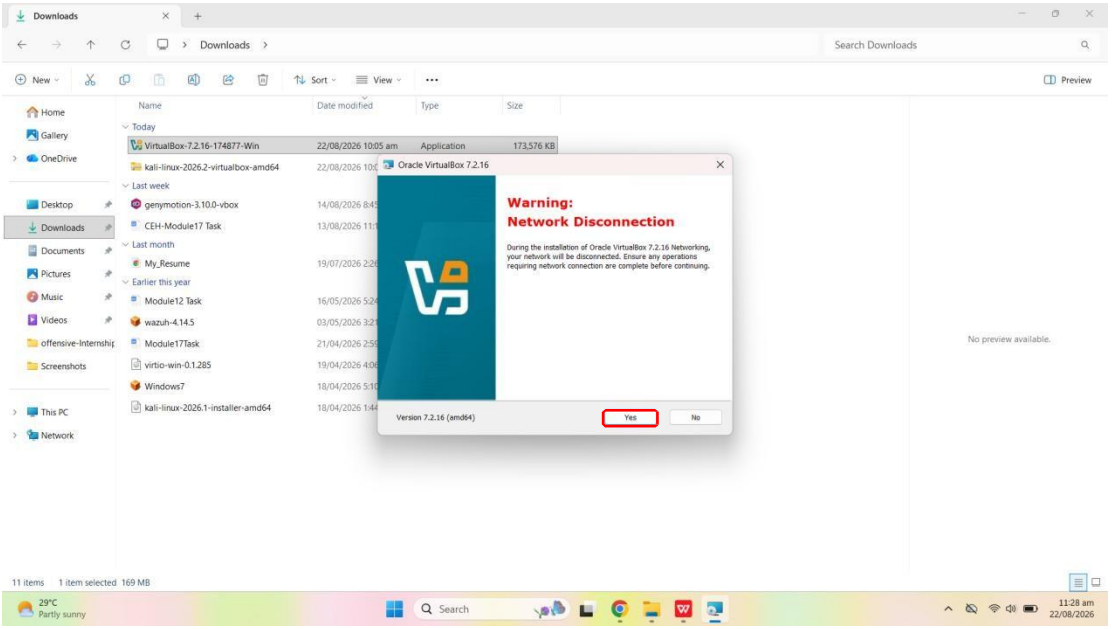
3) Click on I accept the terms of license aggrement and then click next.



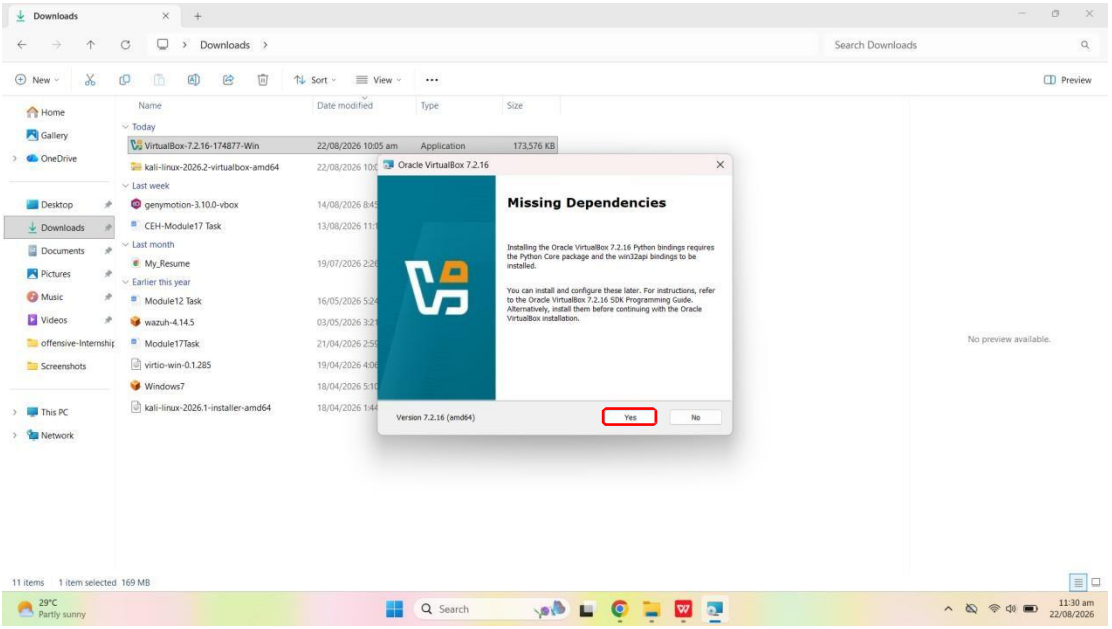
4) As default location of virtual box will be in C drive simply Press next.



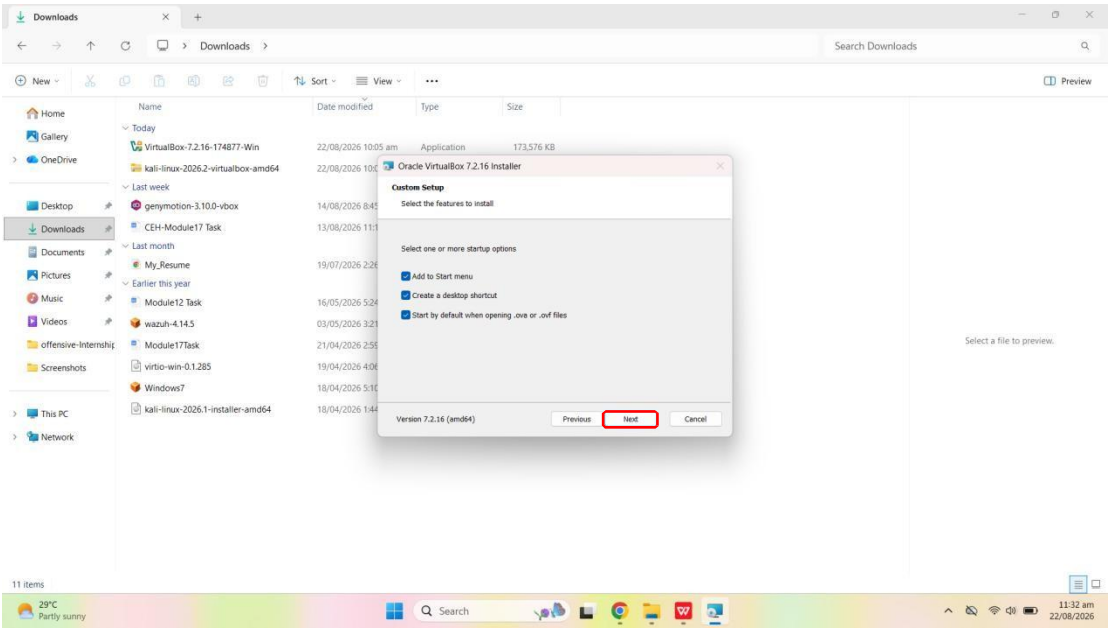
5) During the Installation there may be a slightly network interruption for a while. Click next



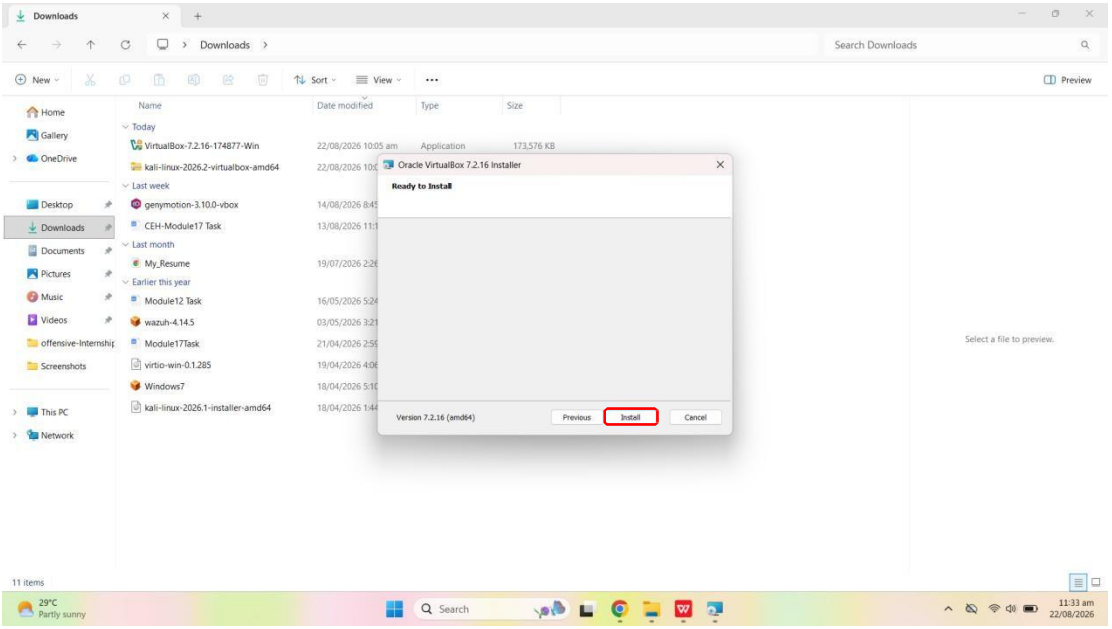
6) Click Yes



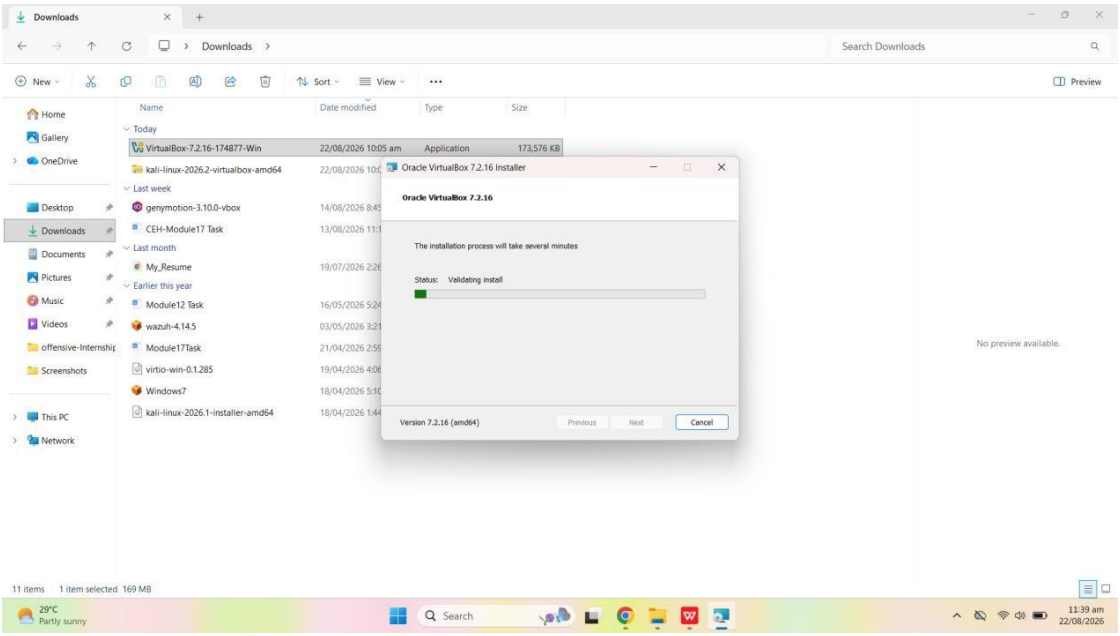
7) Click Next



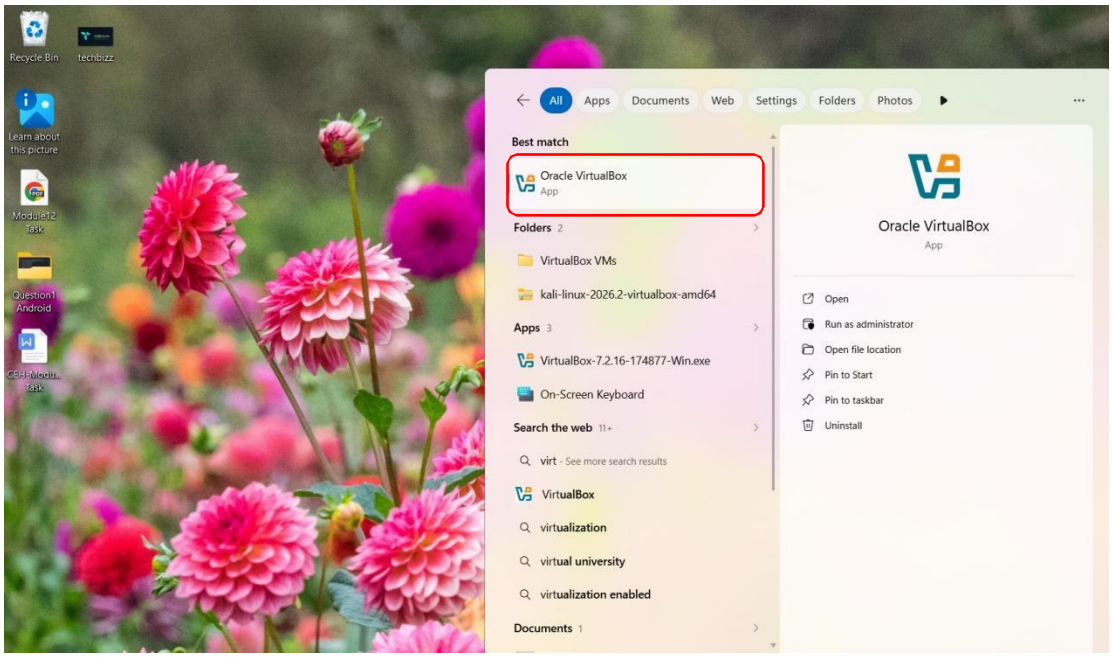
8) Click Install



9) Now installation is in Progress wait for few minutes.



10) Virtual box is successfully downloaded.



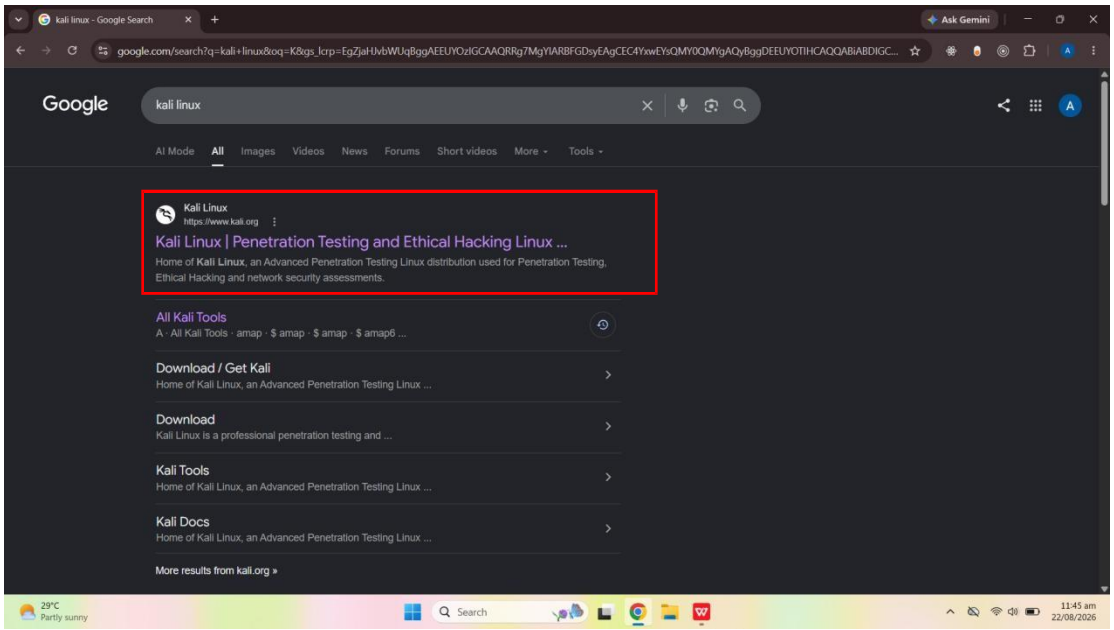
5 Downloading and Installing Kali Linux

5.1 Downloading Kali Linux

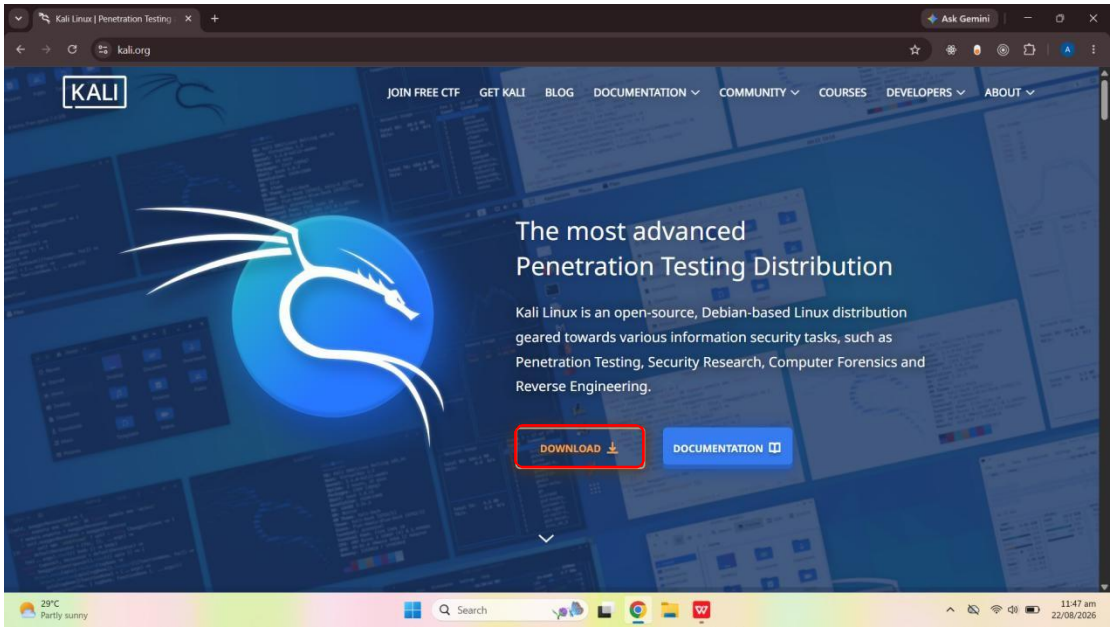
Downloading Kali Linux has 2 options Installer Images and Virtual box Images. Virtual Box Images is a easy way because kali Linux is Pre-built vm means all the configurations are already set by default no need to install and configure anything.

To Download kali linux Follow the steps given below

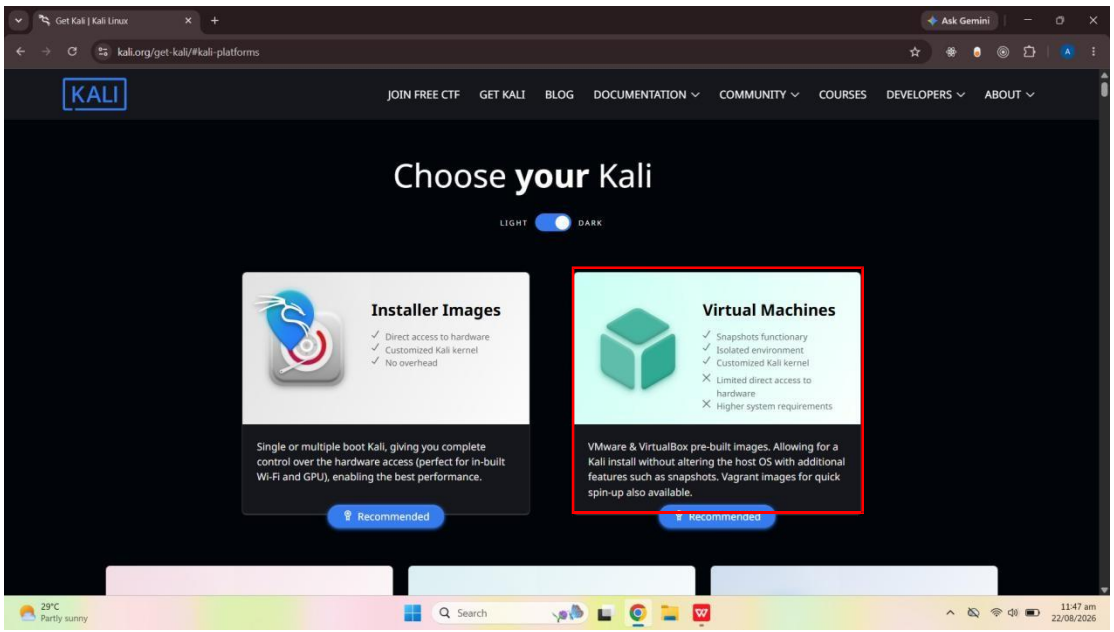
1. Go to google.com and Search Kali Linux select the official Kali linux website shown in Image below. https://www.kali.org



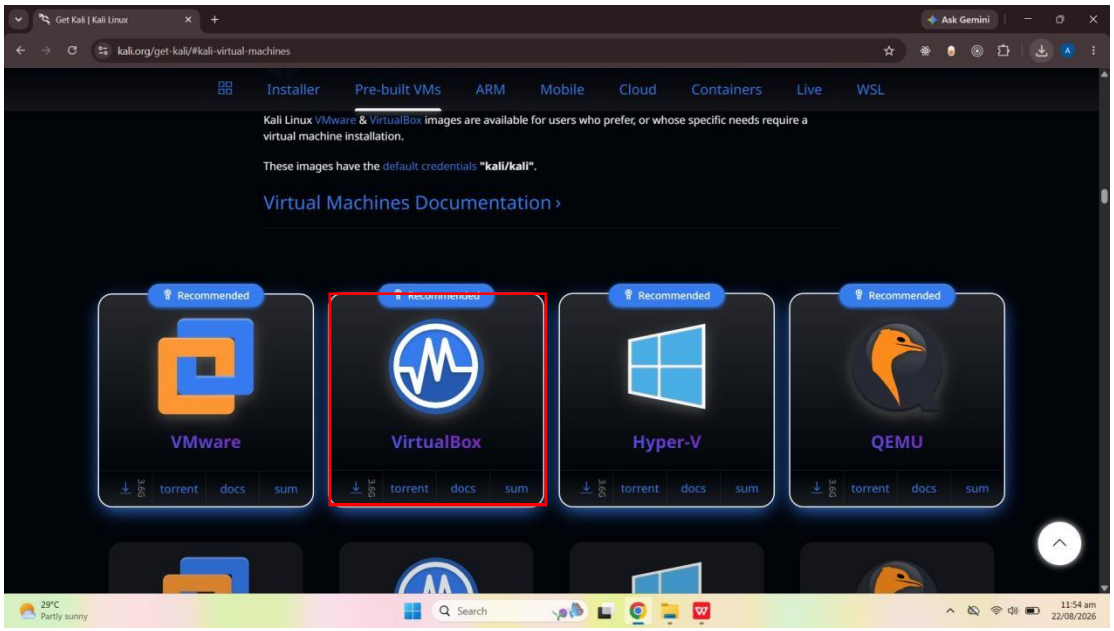
2. After visiting the website click on Download



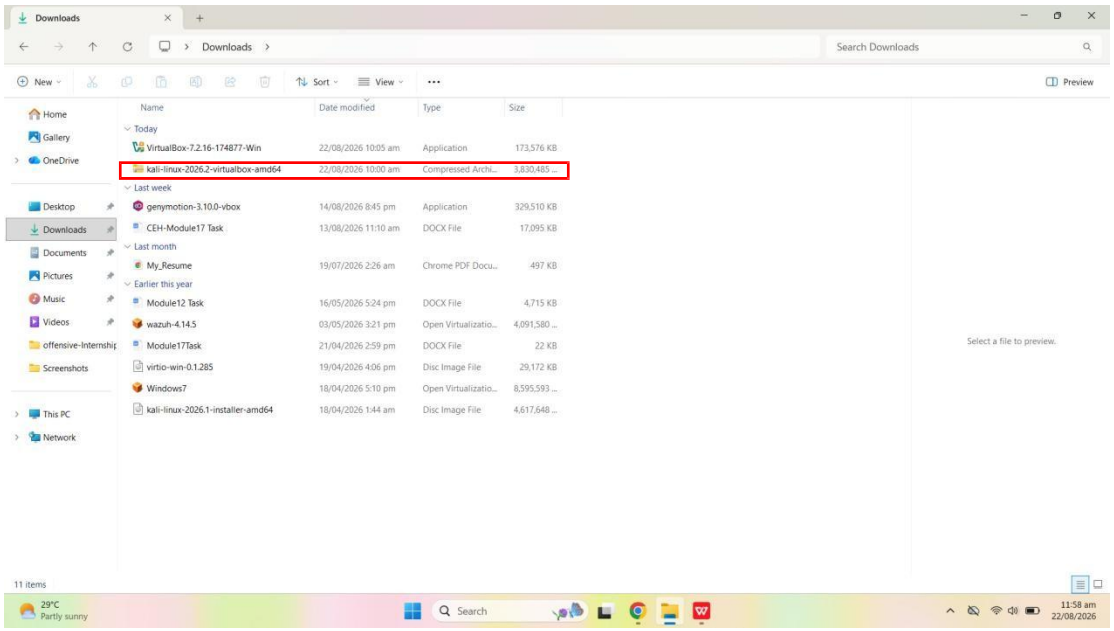
3. Select the Virtual machines.



4. Click on the Virtual box as shown in Image below to download It will take some time depending upon the internet Speed



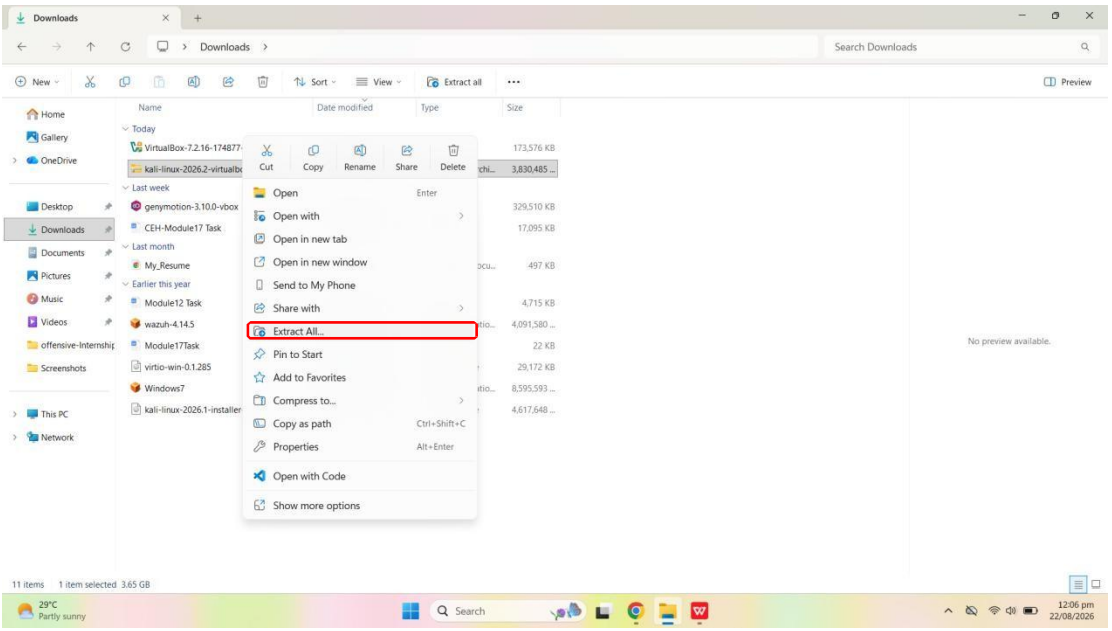
5. As kali Linux Pre-built Vm is downloaded in System Successfully can be shown in Downloads folder.



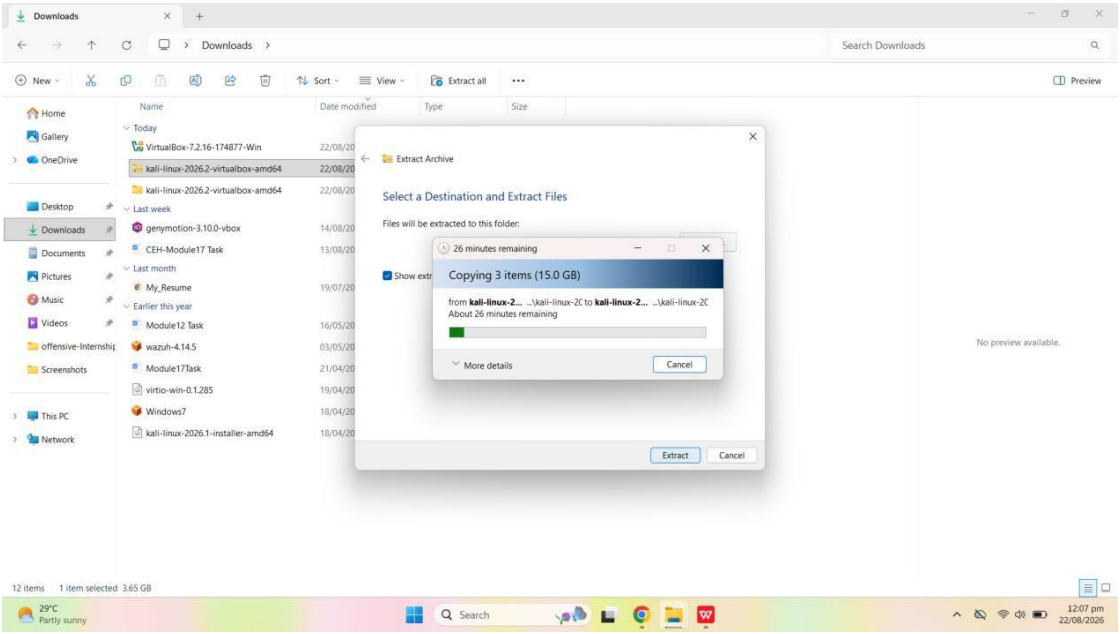
5.2 Installing Kali Linux

To Install Kali linux Follow the steps given below

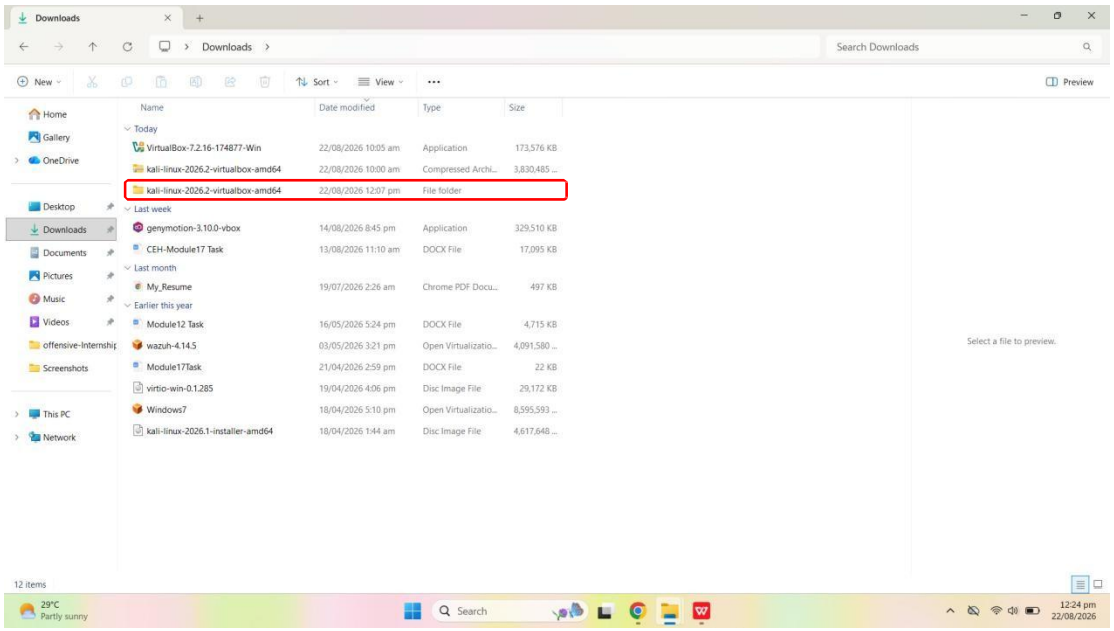
1) Right click on the zipped folder of kali linux and Click on Extract all



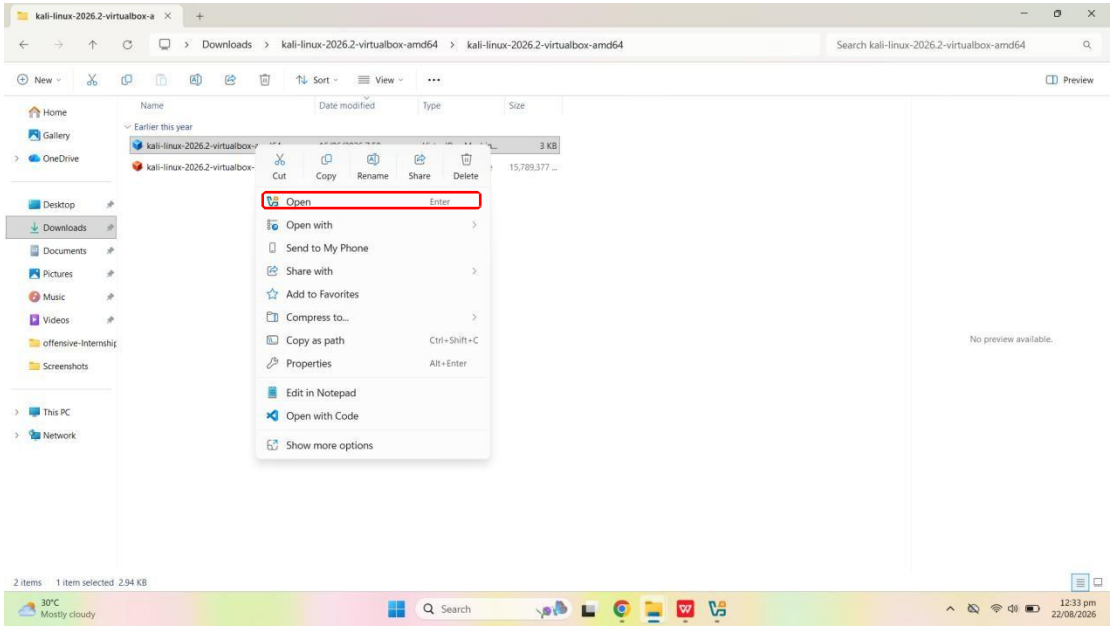
2) Select Destination and then Press Extract Extraction Process is in Continue wait for few minutes



3) As Extraction is completed Successfully



4) Open the Extracted folder and click on file pre built vm of kali linux will be added in virtual box.

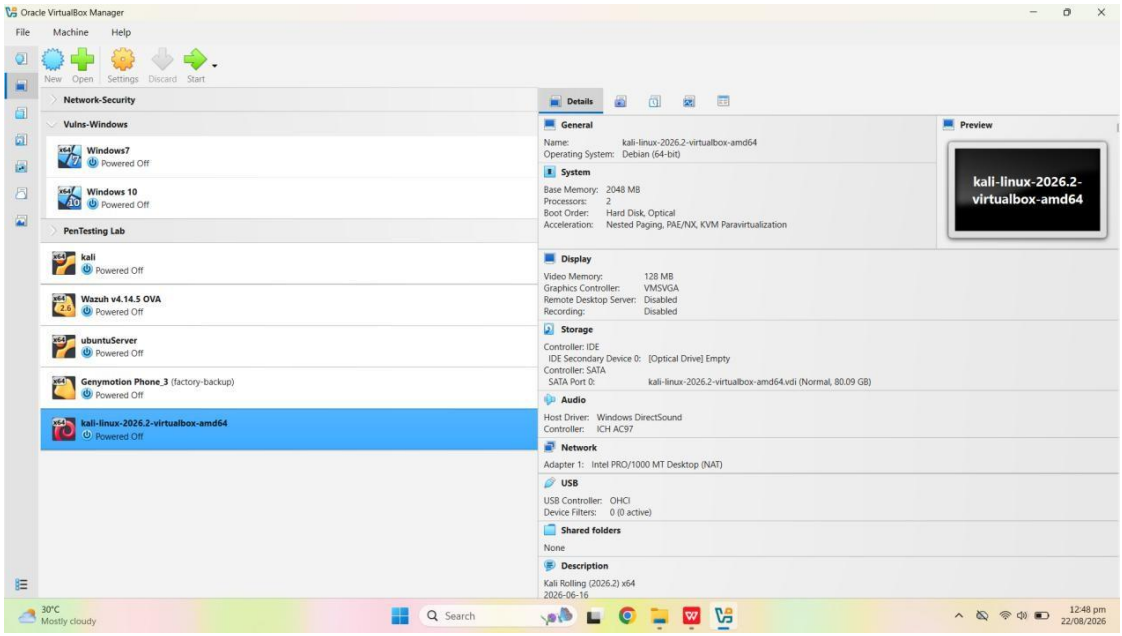


5) Kali Linux machine is successfully imported in oracle virtual box. Now its time to check the assigned hardware resources before booting the kali.

Virtual Machine Configuration

1. **Operating System**: Debian-based
2. **Ram**: 2GB
3. **Processors**: 2 Cpu Cores
4. **Storage**: 80GB
5. **Network Adapter**: NAT

These are the default Configuration of Kali Virtual Machine we can adjust hardware resources as per our System's specification for better performance.



As Kali Linux is successfully bootted up know



Open the Terminal and Run the commands one by one to check user account and Network configuration is working properly Commands are mentioned below

- whoami
- ip a
- sudo apt update

6 Basic Kali Linux Commands

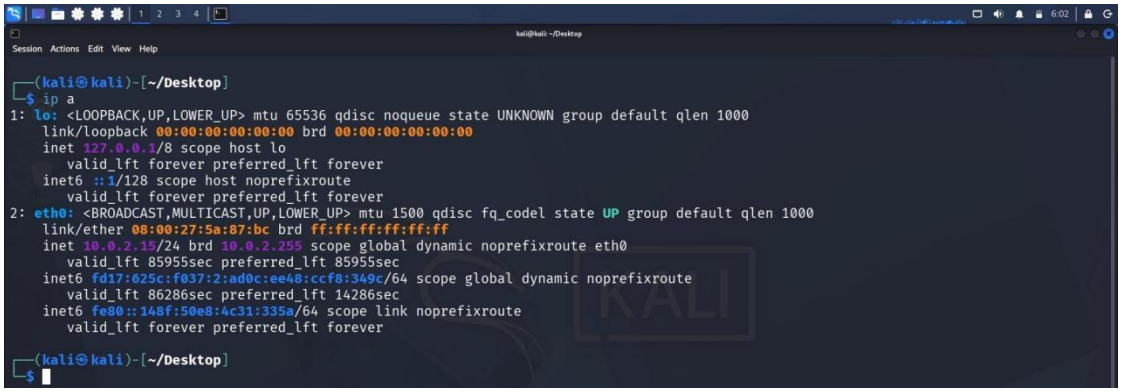
6.1 Command whoami

This command is use to check the current logged-in user.



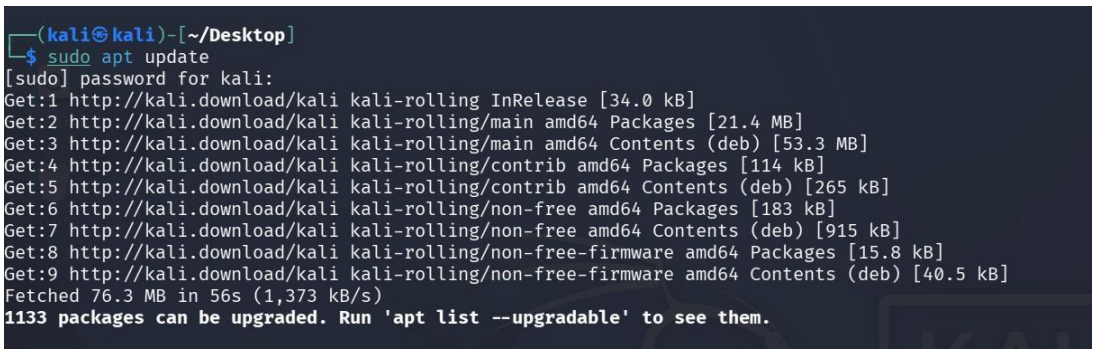
6.2 Command ip a

This command is use to check the ip address of a system.



6.3 Command sudo apt update

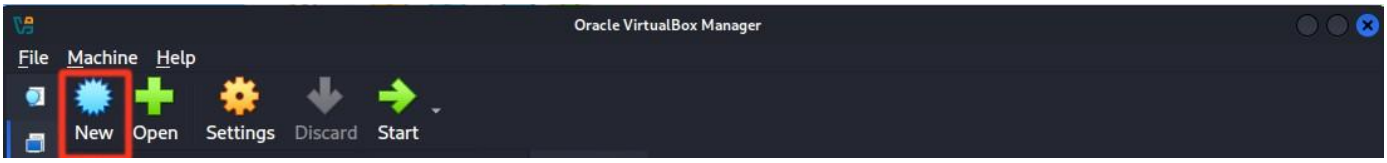
This command is use to update all the packages from the online repositories inside system so every time we get a fresh package it must be run with root privileges as shown in image below. **sudo** is for root user.



7 Installing Windows 11 in virtual box

Step No.1

- Click New Button shown in image below



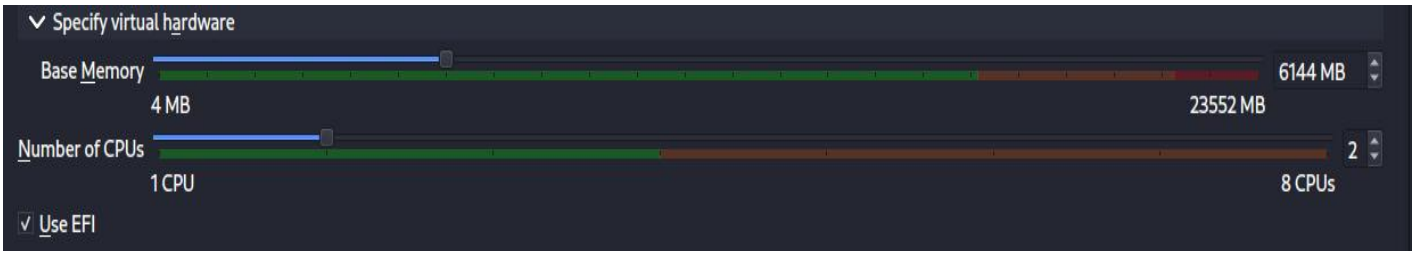
Step No.2

- **VM Name:** Windows 11
- **Select ISO Image:** Windows 11 (No TPM)
- **Select OS:** Microsoft Windows
- **Select OS Version:** Windows 11 (64-Bit)
- Unchecked Proceed with Unattented Installation



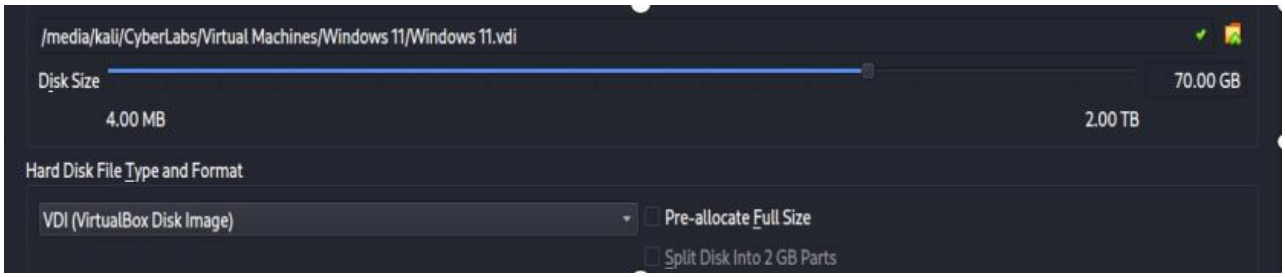
Step No.3

- Conf Base Memory: **6 GB**
- Conf Number of CPUs: **2**



Step No.4

- Disk Size: **70 GB**
- Hard Disk File Type: **VirtualBox Disk Image**
- Unchecked Pre-Allocate Full Size
- Click Finish Button and Start the Machine



Step No.5

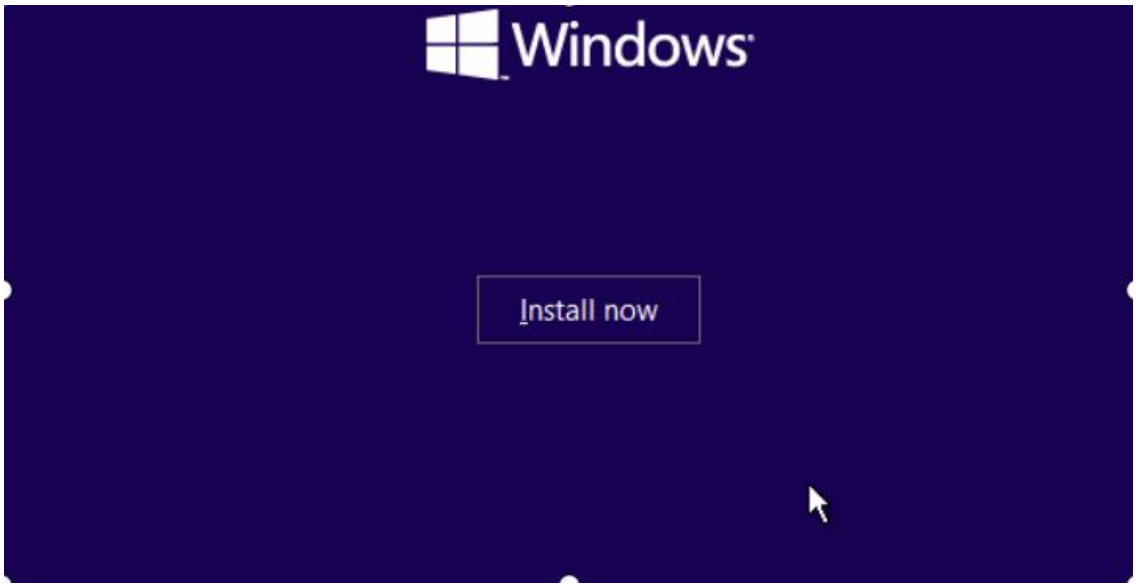
- Select Option: My language is English



Step No.6

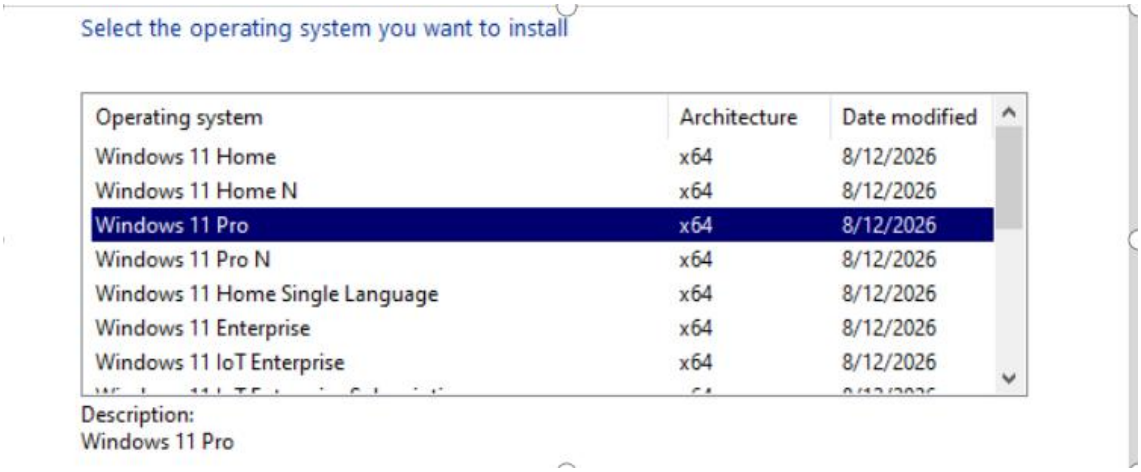
- Click **Next** Button
- Click **Install Now** Button

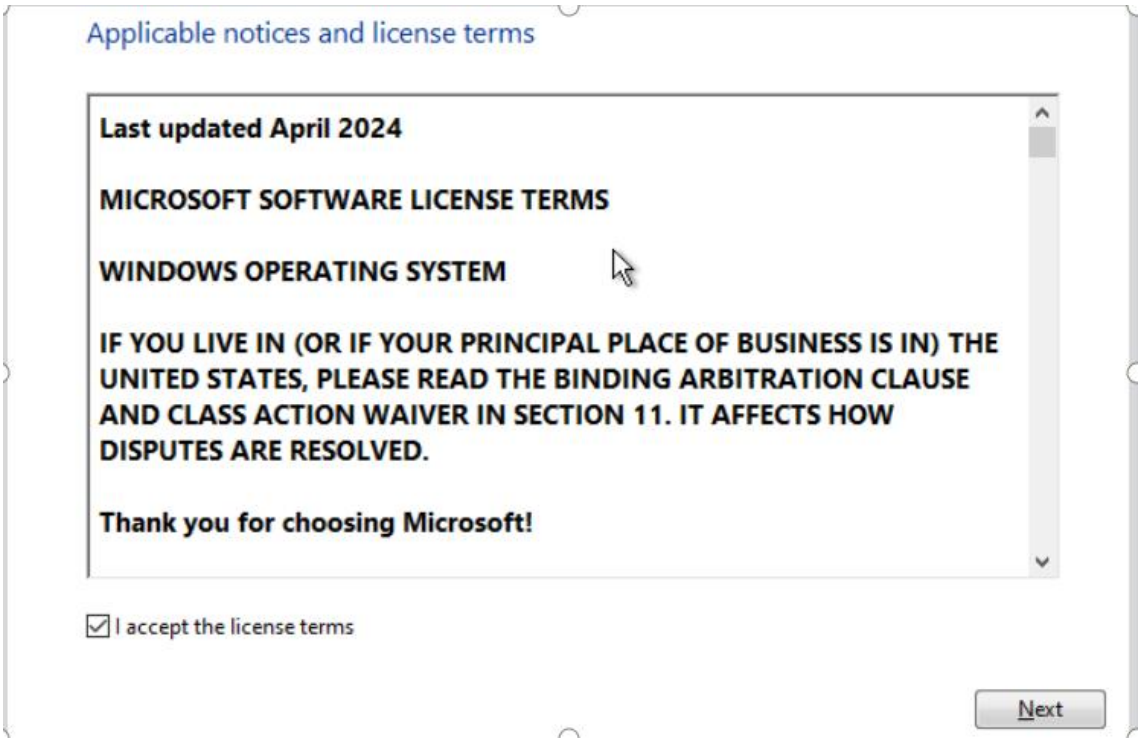




Step No.7

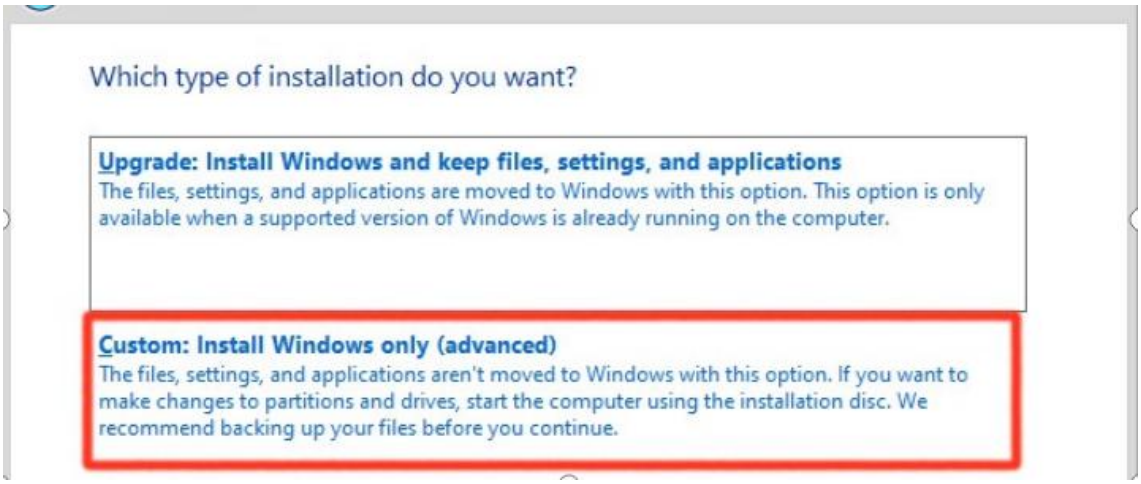
- Select Option: **Windows 11 Pro** and Click **Next** Button
- Checked "I accept the License terms" and Click **Next** Button

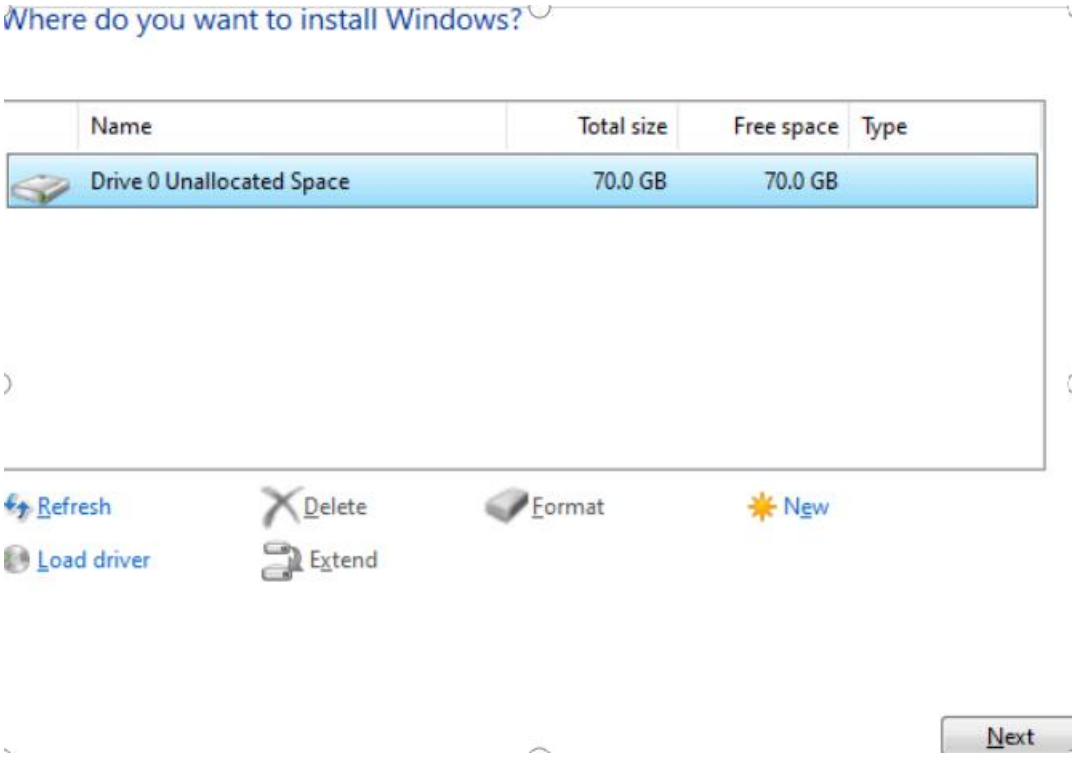




Step No.8

- Click Custom Installation
- Select **Disk** and Click **Next** Button

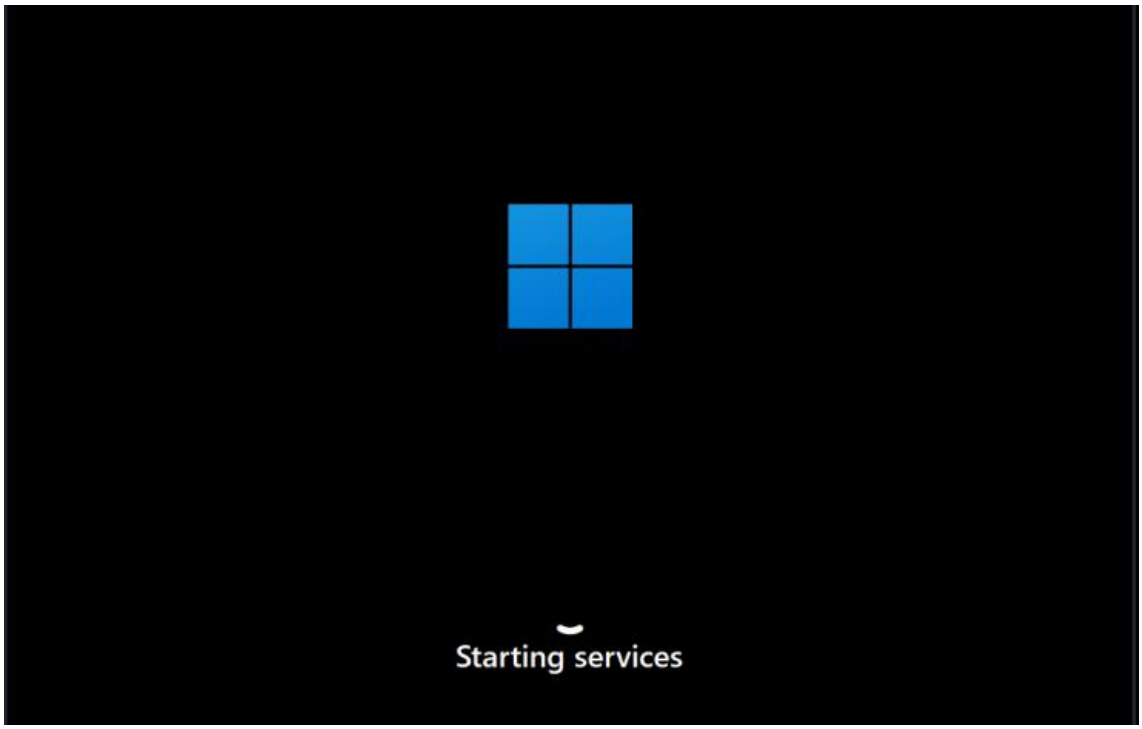




Step No.9

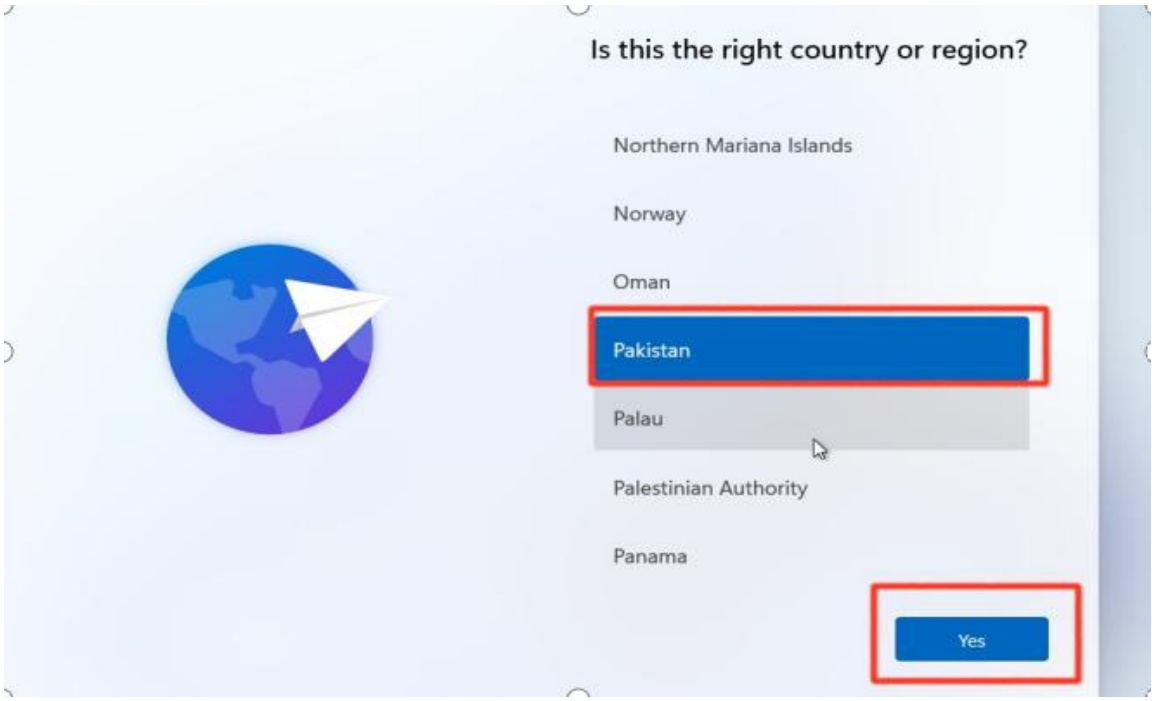
- Installing windows Features and Updates
- Wait a Few Minutes starting window

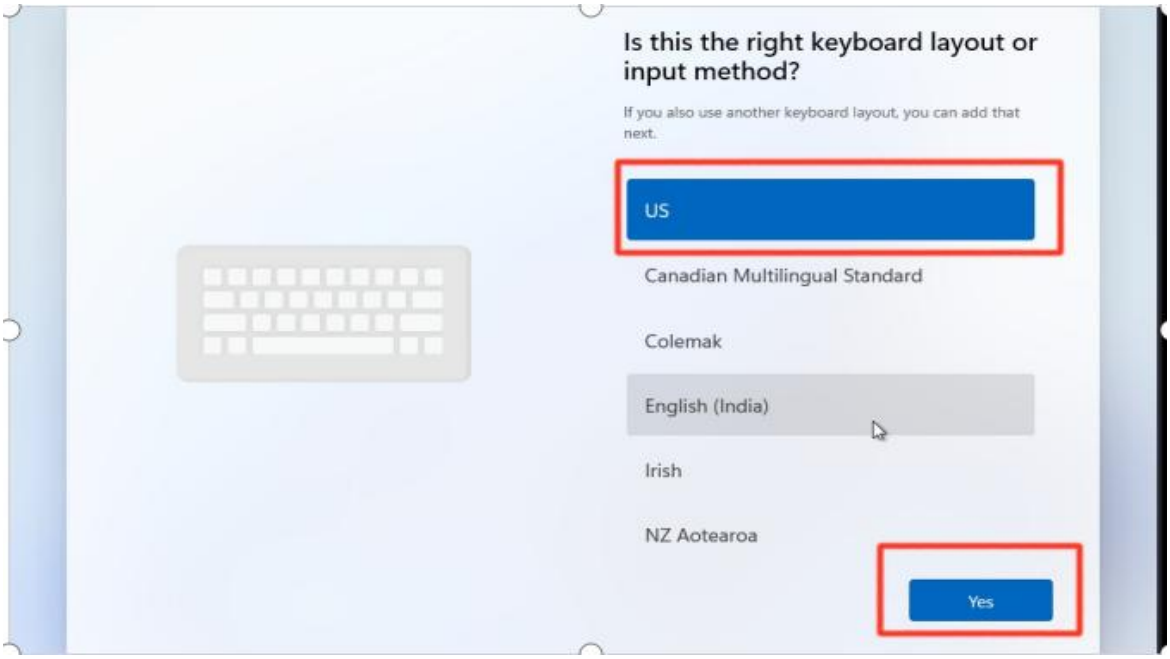




Step No.10

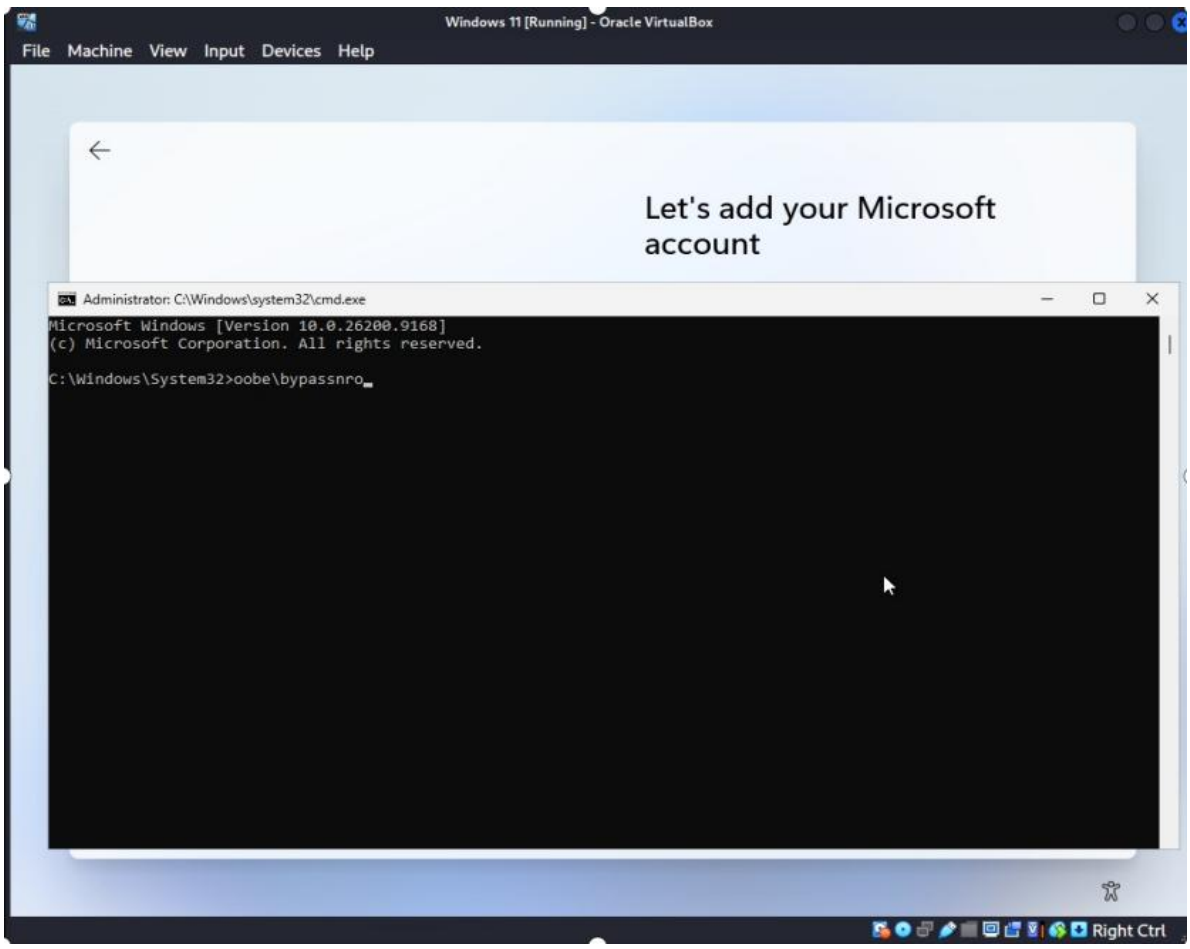
- Select Country: **Pakistan**
- Click **Yes** Button and Select Keyboard Layout or Input Method

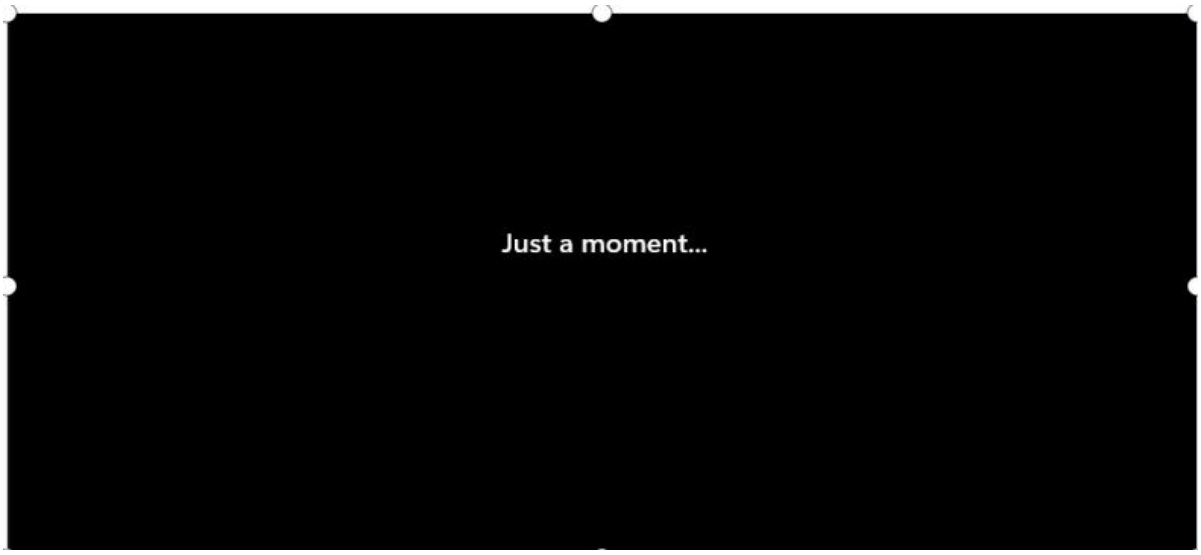




Step No.11

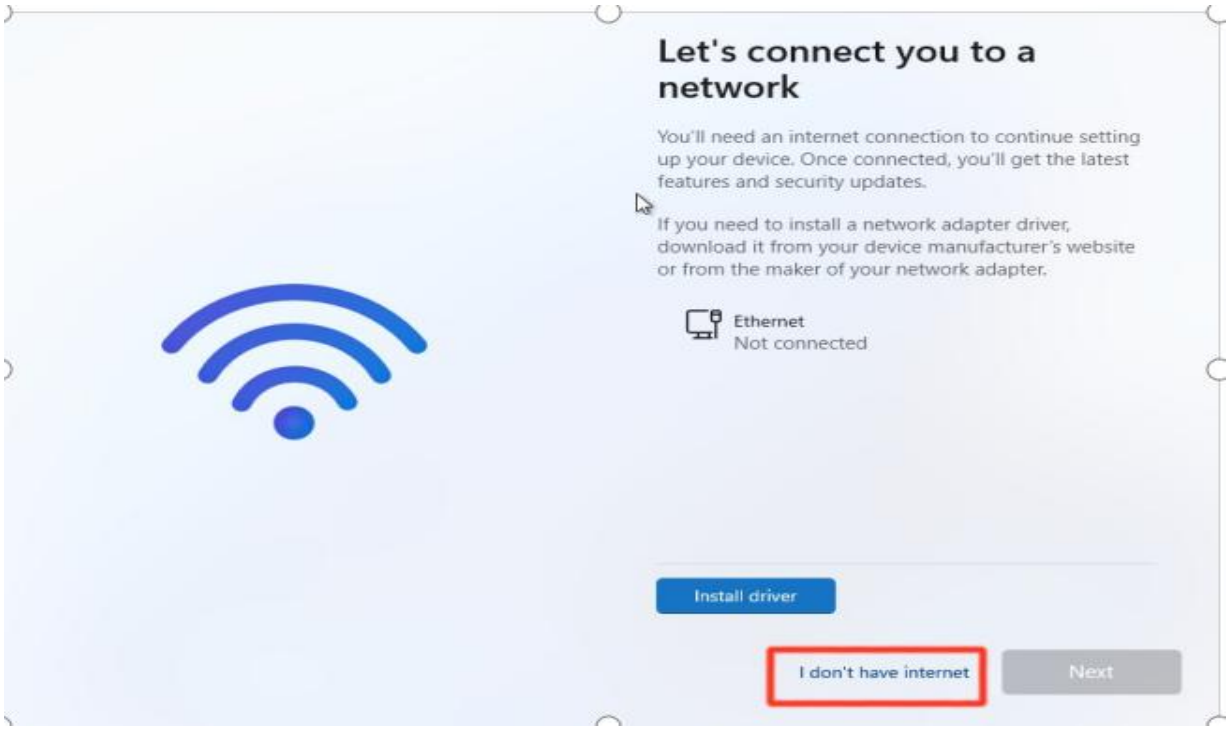
- Press Keyboard Button: **Shift+F10**
- Open CMD and Type This Command: **oobe\bypassnro**
- Wifi Disable and bypass Microsoft Sign in Account.





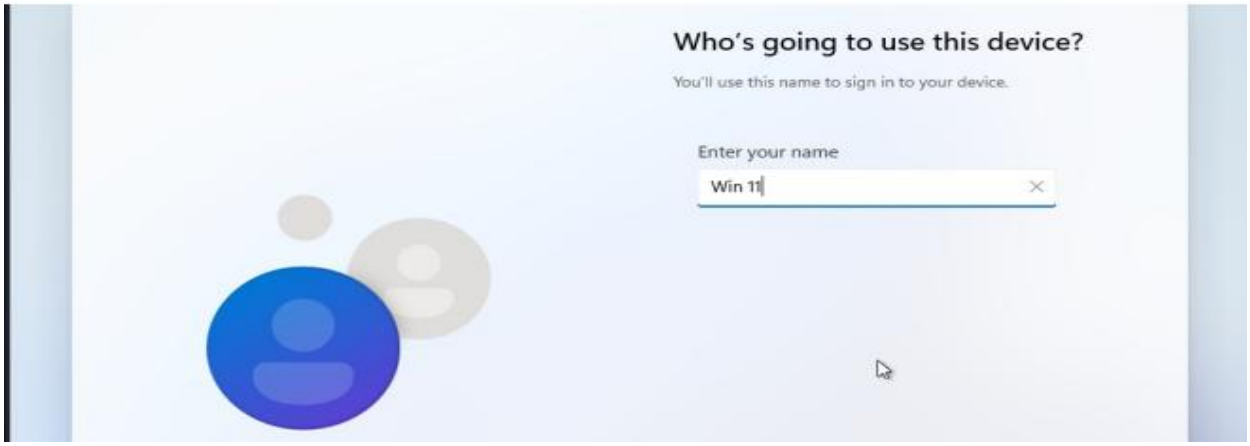
Step No.12

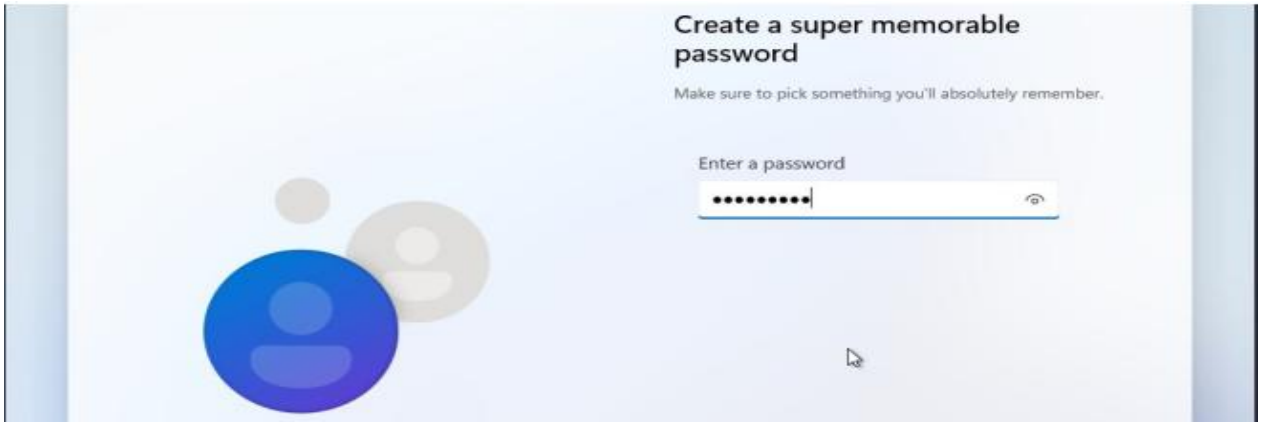
- Click on "I don't have internet"

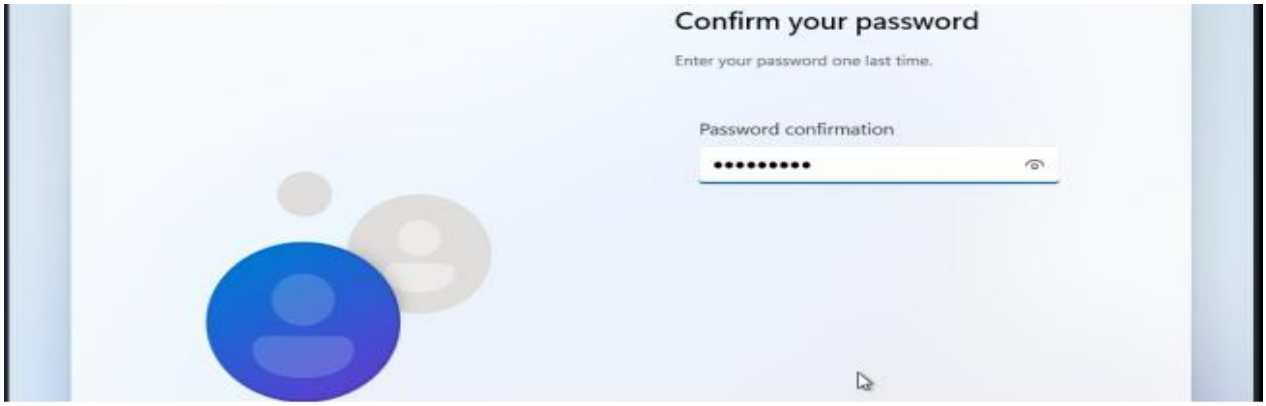


Step No.13

- Conf your Local Account Name: **Win 11** and Click **Next** Button
- Set Password for Account and Confirm Password

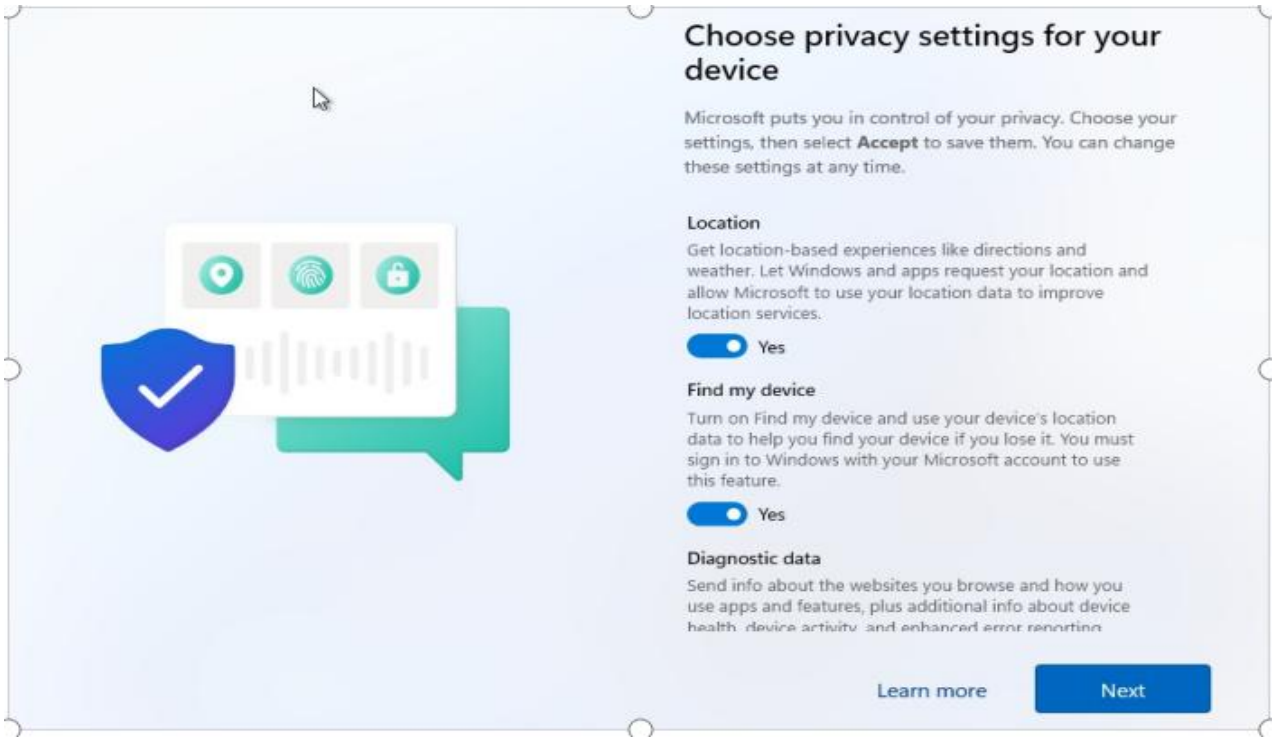


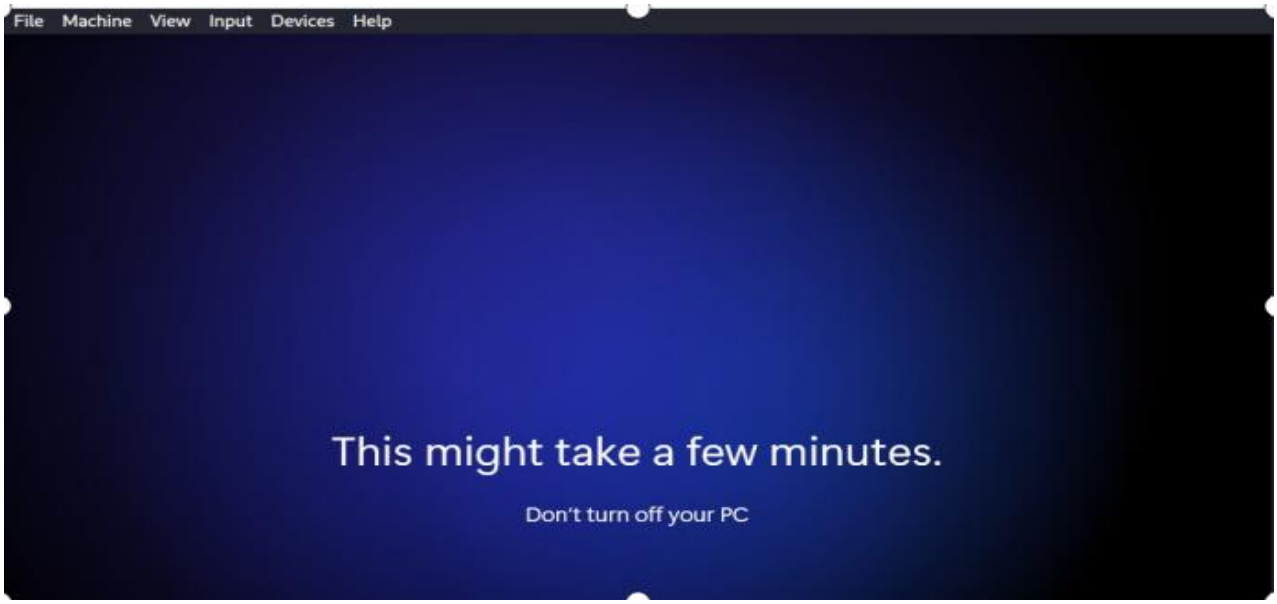




Step No.14

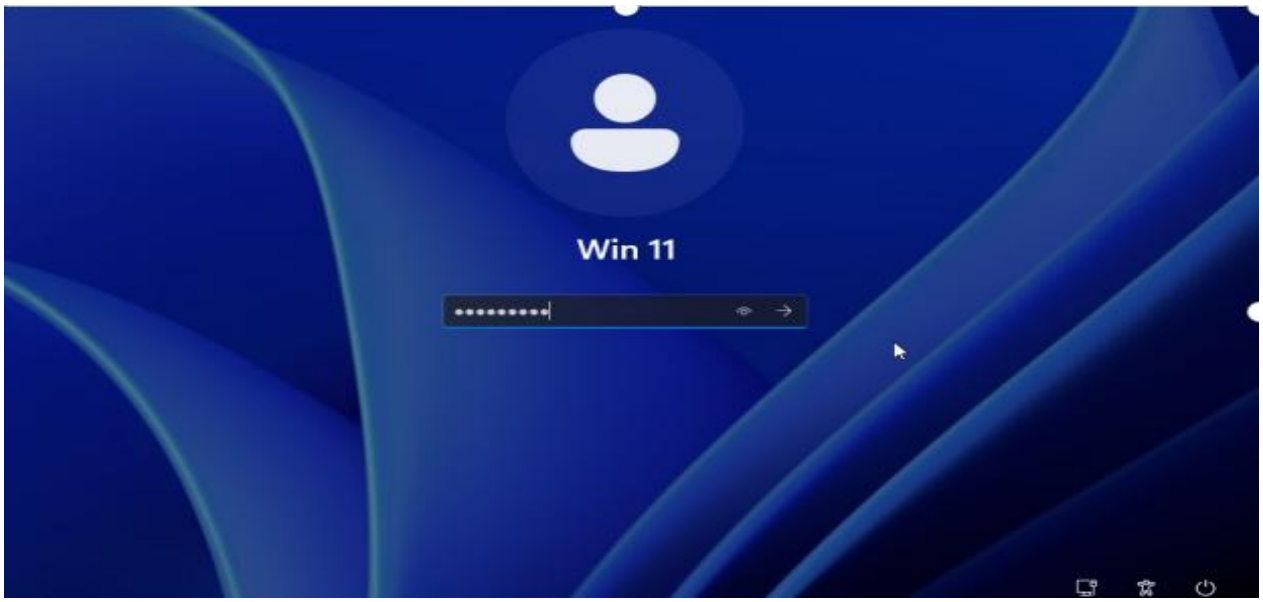
- View Settings and Click **Next** Button

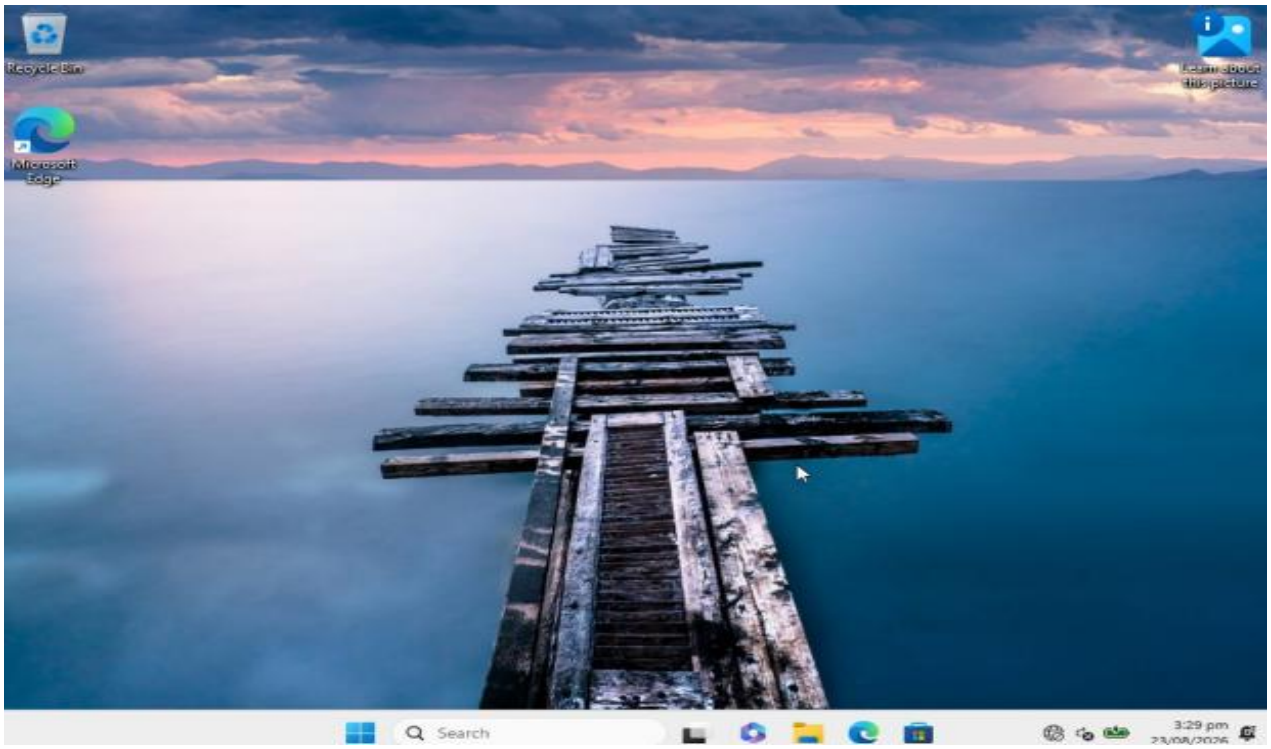




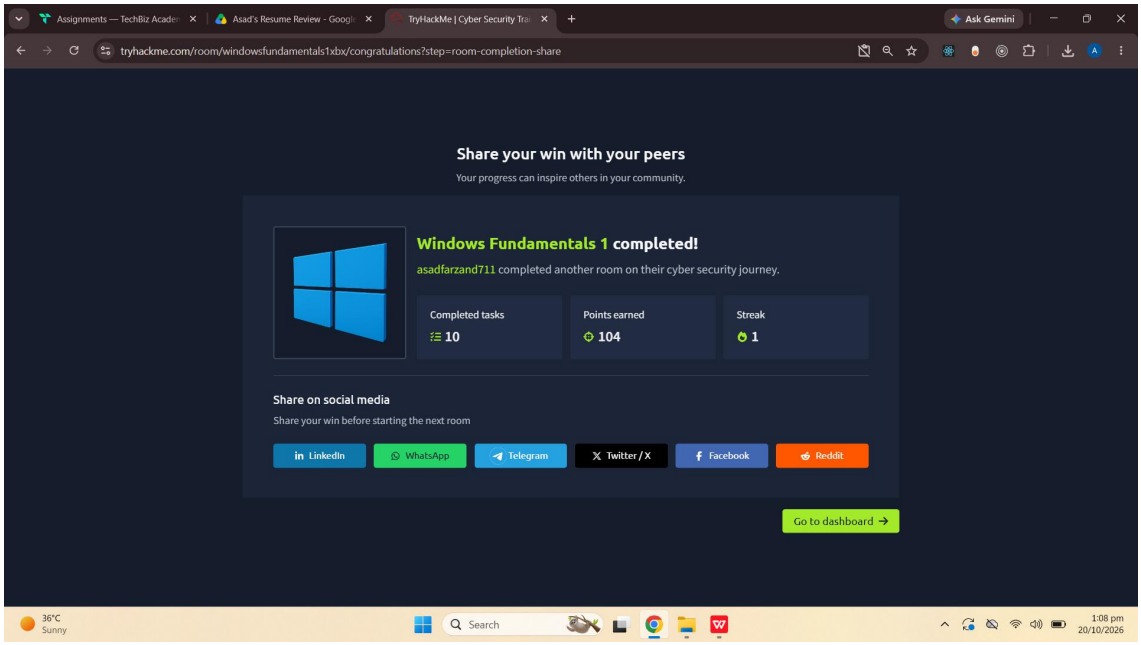
Step No.15

- Enter Password which you Set on **Win 11** Account


Win 11


Recycle Bin
Microsoft Edge
Search
3:29 pm

8 TryHackMe Windows Fundamentals



9 Conclusion

This week I started my Internship from Understanding the core terminologies of cyber security then moved to some networking concepts and then Setting up Isolated and Professional Lab environment for cyber security. This week serves as the base of building strong security skills in upcoming task.